

Chapter 34. Noninvasive Ventilation on a General Ward

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Figure 34-1

Noninvasive Ventilation on a General Ward: Introduction

In any discussion about location of a noninvasive ventilation (NIV) service, it is important to note that the model of hospital care differs between countries and that there may be significant differences even between hospitals within the same country. There will be variations in staffing levels; the skills of doctors, nurses, and paramedical staff; and the sophistication of monitoring. The terms *intensive care unit* (ICU), *high-dependency unit* (HDU), and *general ward* have a different meaning to different people. Care therefore must be taken when extrapolating experience and results obtained in one environment to other hospitals and countries.

The United Kingdom’s King’s Fund panel¹ defines intensive care as “a service for patients with potentially recoverable diseases who can benefit from more detailed observation and treatment than is generally available in the standard wards and departments.” The definition of HDU is less clear, with some HDUs allowing invasive monitoring, whereas in others only noninvasive monitoring is performed. In some countries, specific respiratory ICUs and intermediate ICUs have been developed.^{2,3} Specifically, within the King’s Fund definition is the consideration of intensive care as a service rather than a place; critical care is provided within a continuum of primary, secondary, and tertiary care, and patients are categorized on the basis of their needs⁴ (Table 34-1). Movement through the different levels usually means transfer from one location to another. Critical care outreach teams can advise on care as patients cross organizational boundaries and also facilitate transfer when this is needed.^{5,6} Although most acute NIV services are situated in a specific clinical area, a peripatetic model has been described and has some advantages.^{7,8}

Table 34-1: Classification of Individual Patient Dependency

<i>Level 0:</i> Patients whose needs can be met through normal ward care in an acute hospital
<i>Level 1:</i> Patients at risk of their condition deteriorating or patients recently relocated from higher levels of care whose needs can be met in an acute ward with additional advice and support from the critical care team
<i>Level 2:</i> Patients requiring more detailed observation or intervention, including support for a single failing organ system or postoperative care, and those stepping down from higher levels of care
<i>Level 3:</i> Patients requiring advanced respiratory support alone or basic respiratory support, together with the support of at least two organ systems. This level includes all complex patients requiring support for multiorgan failure

For the purposes of this chapter, the following definitions are used:

- Intensive care.* High ratio of staff to patients, facility for invasive ventilation and sophisticated monitoring.
- Intermediate respiratory ICU or HDU.* Continuous monitoring of vital signs, with a staffing ratio intermediate between an ICU and a general ward, in a specified clinical area. Intubated patients (unless with tracheostomy) usually are not cared for in this environment.
- General ward.* Takes unselected emergency admission, and although most wards will have a particular speciality interest, it is likely that because of the unpredictability of demand, patients with a variety of conditions and degrees of severity will be cared for in the same clinical area. Nurse staffing levels vary, but the intensity of nursing input available in HDUs and ICUs is not possible. Only basic monitoring is available.

Another important issue when considering NIV in different locations is the severity and acuteness of the insult leading to ventilatory failure. Ventilatory failure can be considered *acute* when it occurs on a background of normal function, *acute-on-chronic* when there is a sudden deterioration on a background of impaired function, or *chronic* when there is ventilatory failure but with no precipitating acute event. Assisted ventilation can be considered *necessary* when without it death will ensue over a few hours or *desirable* when the primary aim is to improve quality of life and also to improve survival over the longer term.

Evidence and Rationale for Noninvasive Ventilation on a General Ward

For additional information about noninvasive ventilation, see [Chapter 18](#).

Acute Noninvasive Ventilation

Chronic Obstructive Pulmonary Disease

NIV first became established as a viable technique for patients with acute respiratory failure secondary to an exacerbation of chronic obstructive pulmonary disease (COPD) in the ICU. The most striking finding from the early randomized, controlled trials (RCTs) comparing NIV with conventional therapy was a reduction in the need for intubation,^{9,10} which in the largest study translated into improved survival and reduced length of both ICU and hospital stays.⁹ Complications, particularly pneumonia and other infectious complications, were reduced markedly.^{9,11–15} It is striking that NIV was administered for only a relatively small proportion (mean: 6 hours) of each day⁹ or at modest levels for a longer period.¹⁰ With NIV, paralysis and sedation are not needed, and ventilation outside the ICU is an option. Given the considerable pressure on ICU beds in some countries, the high costs, and that for some patients admission to ICU is a distressing experience,¹⁶ this is an attractive option.

There have been seven prospective, randomized, controlled studies of NIV outside the ICU either on general wards, in an intermediate unit, or in the emergency department.^{17–23} A more rapid improvement in abnormal physiology is a consistent finding, but it was only in the largest,²² adequately powered, study that a benefit in terms of outcome was seen. Plant et al²² recruited 236 patients with an acute exacerbation of COPD, who were still hypercapnic, with a pH less than 7.35, and respiratory rate greater than 23 breaths/min on arrival on the ward. A proportion of patients will improve just with medical therapy. In a 1-year-period prevalence study²⁴ of patients with acute exacerbations of COPD, 20% of 954 patients were acidotic on arrival in the emergency department; of these, 25% had completely corrected their pH by the time of arrival on the ward. There was a weak relationship between partial pressure of arterial oxygen (PaO₂) on arrival at hospital and the presence of acidosis, suggesting that, in at least some patients, respiratory acidosis had been precipitated by high-flow oxygen therapy administered on the way to hospital.

The study was performed on general respiratory wards in thirteen centers. NIV was applied, by the usual ward staff, using a bilevel device in spontaneous mode according to a simple protocol. “Treatment failure,” a surrogate for the need for intubation, defined by a priori criteria, was reduced from 27% to 15% by NIV. In-hospital mortality was reduced from 20% to 10%. This study suggests that with adequate staff training, NIV can be applied with benefit outside the ICU by the usual ward staff and that early introduction of NIV in a general ward results in better outcomes than providing no ventilator support for acidotic patients outside the ICU. A recent national audit in the United Kingdom,²⁵ however, raised significant concerns about the provision of NIV in the “real” world. Although it was not recorded in the audit, it is likely that the majority of patients received NIV outside of ICUs, mostly on general wards. Two hundred and thirty-two hospital units collected data on 9716 patients of whom 1077 received NIV. Of concern 30% of patients with persisting respiratory acidosis did not receive NIV. The mortality was higher in all acidotic groups receiving NIV than in those treated without. Patients who had late-onset acidosis had a particularly poor prognosis confirming the results of an earlier case series.²⁶ Interestingly, 11% of acidotic admissions had a pure metabolic acidosis. There is a challenge in translating the results of RCTs into everyday clinical practice, especially when the particular technique involves significant technical expertise. It reinforces the need for ongoing audit to ensure that standards are maintained.

Conditions Other Than Chronic Obstructive Pulmonary Disease

Trials in acute exacerbations of COPD provide the biggest body of evidence on NIV. NIV, however, is also used in other conditions, often on the basis of what has been learned in COPD.

Hypoxemic and Hypercapnic Respiratory Failure.

Obesity.

Obese patients may present with acute or acute-on-chronic respiratory failure. In numerical terms this patient group is increasing; the number of patients requiring home ventilation because of obesity-hypoventilation syndrome is increasing year on year, and in one study, patients with obesity now comprise the largest single group.²⁷ For obese patients requiring ventilator support acutely, the outcome from invasive ventilation is generally poor.²⁸ There are major practical problems associated with nursing critically ill obese patients, often requiring many pairs of hands and specialized lifting equipment for basic tasks. There are no RCTs of the use of NIV in patients with ventilatory failure secondary to obesity. A case series in which patients who received NIV were compared with those who refused it showed a survival advantage for those receiving NIV (97% vs. 42%)²⁹; this was not controlled and there may have been other reasons for the difference. Very obese patients may have upper airway obstruction during sleep and because the impedance to inflation may be very high may require different ventilator modes.³⁰

Neuromuscular Disease and Chest Wall Deformity.

Patients with acute respiratory failure secondary to neuromuscular disease and chest wall deformity are not widely studied because they are small in number. Because of markedly reduced respiratory reserve, however, these patients are often challenging to wean from invasive ventilation and endotracheal intubation is best avoided if possible. Ideally, at-risk patients should already be under follow-up in a specialist unit and have been warned of the symptoms of evolving respiratory failure and of the necessity to present to hospital early in case of changes in their condition. Some patients will already have experienced a trial of domiciliary NIV. As such they represent good candidates for NIV outside the ICU. In addition to staff skilled in the delivery of NIV, therapists with expertise in secretion clearance techniques, including the use of mechanical insufflators or exsufflators,^{31,32} are vital in the management of these patients.

Cardiogenic Pulmonary Edema.

Cardiogenic pulmonary edema (CPE) represents a special case because the onset and recovery are usually both rapid. Most patients present to the emergency room, but some develop CPE in the ward. There have been seven systematic reviews (meta-analyses) on noninvasive ventilator assistance in CPE published since 2005.^{33–40} Overall, there was a significant reduction in mortality for those patients treated with continuous positive airway pressure (CPAP) and a trend toward improved survival with NIV.³⁴ Both CPAP and NIV showed benefit when intubation was an outcome. There was no difference in any outcome when CPAP and NIV were compared. There was a trend toward an increase in myocardial infarction rate with NIV, but this was largely caused by the weighting of one study.⁴¹ Two recent trials may result in the reappraisal of the role of NIV in acute CPE.^{42,43}

In the 3CPO trial,⁴² a multicenter, open, prospective RCT, patients were randomized to standard oxygen therapy, CPAP, or bilevel ventilation. There was no difference between 7-day mortality for standard oxygen therapy (9.8%) and NIV (CPAP and bilevel ventilation, 9.5%; $P = 0.87$). The combined end point of 7-day death or intubation rate was similar irrespective of NIV modality (11.7% vs. 11.1%, CPAP vs. bilevel ventilation respectively; $P = 0.81$). In comparison to standard oxygen therapy, NIV was associated with greater reductions in breathlessness scores, heart rate, acidosis, and hypercapnia at 1 hour. There were no treatment-related adverse events. There were no differences in other secondary outcomes, such as myocardial infarction rate, intubation, length of hospital stay, or ICU admission rate.

In another trial,⁴³ 120 patients were enrolled in three French emergency departments to either CPAP or NIV. There was no difference between interventions for any outcome. Respiratory distress and physiology improved in both arms. Only 3% of patients required intubation and one died within the first 24 hours.

These outcomes are different from the outcomes in the above meta-analyses, despite similar improvements in physiologic and gas exchange variables. The 3CPO trial was adequately powered and recruited more patients than the total of all the studies included in the meta-analyses. The discrepancy between results from one large, multicenter RCT and previous pooled data are not unique and the limitations of meta-analysis are well known.⁴⁴ Individual trials were composed of small treatment group sizes that varied between nine

and sixty-five patients with recruitment rates of only 10% to 30% (compared to 62% randomized in the 3CPO trial). In the meta-analyses, the small total number of outcome events was well below the recommended threshold of 200,⁴⁵ limiting the generalizability of the findings.

The 3CPO trial may have failed to reveal a difference because the intervention was ineffectively delivered. Mean pressures for both CPAP (10 cm H₂O) and noninvasive positive pressure ventilation (IPAP 14/EPAP 7 cm H₂O) are comparable with previous studies, and improvements in physiologic variables are similar. There was crossover between interventions in all three arms of the 3CPO trial and these were analyzed on an intention-to-treat basis. There were differing reasons with respiratory distress and hypoxia being more likely in the control arm and lack of patient tolerance in the two intervention arms. After these patients were removed from primary outcome analysis, there remained no significant difference between groups, although mortality rates were lower.

Previous trials have indicated that the physiologic improvement seen with NIV is translated into a reduction in tracheal intubation rates.^{33,34} In contrast, the 3CPO trial found no benefit in reducing intubation rates by NIV. Reasons for this are unclear but may reflect the differing patient populations, concomitant therapies, and thresholds for intubation and mechanical ventilation. Intubation rates in the standard therapy arms vary from 35% to 65% in early trials to 5% to 7% for recent trials in emergency department settings, despite similar severity of illness. Intubation rate in the intervention arms have fallen considerably over time, with some initial trials reporting intubation rates of up to 35% whereas recent reports have consistently suggested rates of around 5%. The recent trial⁴³ from France reported a 3% intubation rate, almost identical to that in the 3CPO trial. It is difficult to make direct comparisons because studies differ in the time at which mortality is recorded, but, if anything, survival has probably improved as intubation rates have fallen, suggesting better overall management.

One danger of NIV is that other aspects of medical therapy may be forgotten because the focus is on the application of NIV. Nitrates are key and the total dose delivered has been shown to be an important predictor of outcome.^{46,47} Positive pressure is beneficial to the failing heart and has some similarities to the effects of nitrates (preload and afterload reduction); if medical management is suboptimal, ventilation will have a beneficial effect on the failing heart, which may be lost if these effects have already been achieved with medication.

Finally, the patients recruited may have been less unwell than those in other studies. There was no difference in survival between recruited and nonrecruited patients, and no interaction with disease severity making this unlikely. The physiologic disturbance in these patients put them at the sickest end of the spectrum of patients studied, and, indeed, in contrast to other studies, acidosis (mean pH: 7.22) and hypercapnia (mean partial pressure of arterial carbon dioxide [PaCO₂]: 7.6 kPa) were invariable.

Despite these negative findings, a reduction in dyspnea, which was very intense, was a striking feature in patients receiving ventilator support, and this alone is sufficient reason to utilize ventilation in CPE. There is a trade-off between the beneficial effects of this reduction in dyspnea against discomfort from the mask and other factors.

Hypoxemic Respiratory Failure.

There are no RCTs of NIV outside the ICU in hypoxemic respiratory failure. An RCT in the ICU¹² showed that patients receiving NIV had significantly lower rates of serious complications, and those treated successfully with NIV had shorter ICU stays. Post hoc analysis of patients grouped according to the Simplified Acute Physiology Score (SAPS) showed that NIV was superior to conventional mechanical ventilation in patients with a SAPS less than 16. In patients with a SAPS equal to or greater than 16, outcome was similar irrespective of the type of ventilation. Another study,⁴⁸ in immunocompromised patients, introduced NIV at a much lower level of physiologic compromise than would be required for invasive ventilation, and the sequential strategy (predefined periods on and off NIV) suggests that these patients could manage periods of spontaneous breathing safely. Further data are needed but it is reasonable for selected patients to have a trial of NIV in an experienced noninvasive unit outside the ICU; rapid access to intubation and mechanical ventilation must be available.

Elective Ventilation for Chronic Ventilatory Failure

This subject is dealt with in more detail in [Chapters 28](#) and [33](#). In summary, there is no prospective RCT evidence to support the chronic use of NIV in any patient group. Most practitioners, however, would consider it unethical not to offer NIV to patients with chest wall deformity and neuromuscular disease, and it is unlikely that there will ever be any RCTs of NIV in these conditions. Chronic NIV is not appropriate for most patients with COPD; RCTs are ongoing.

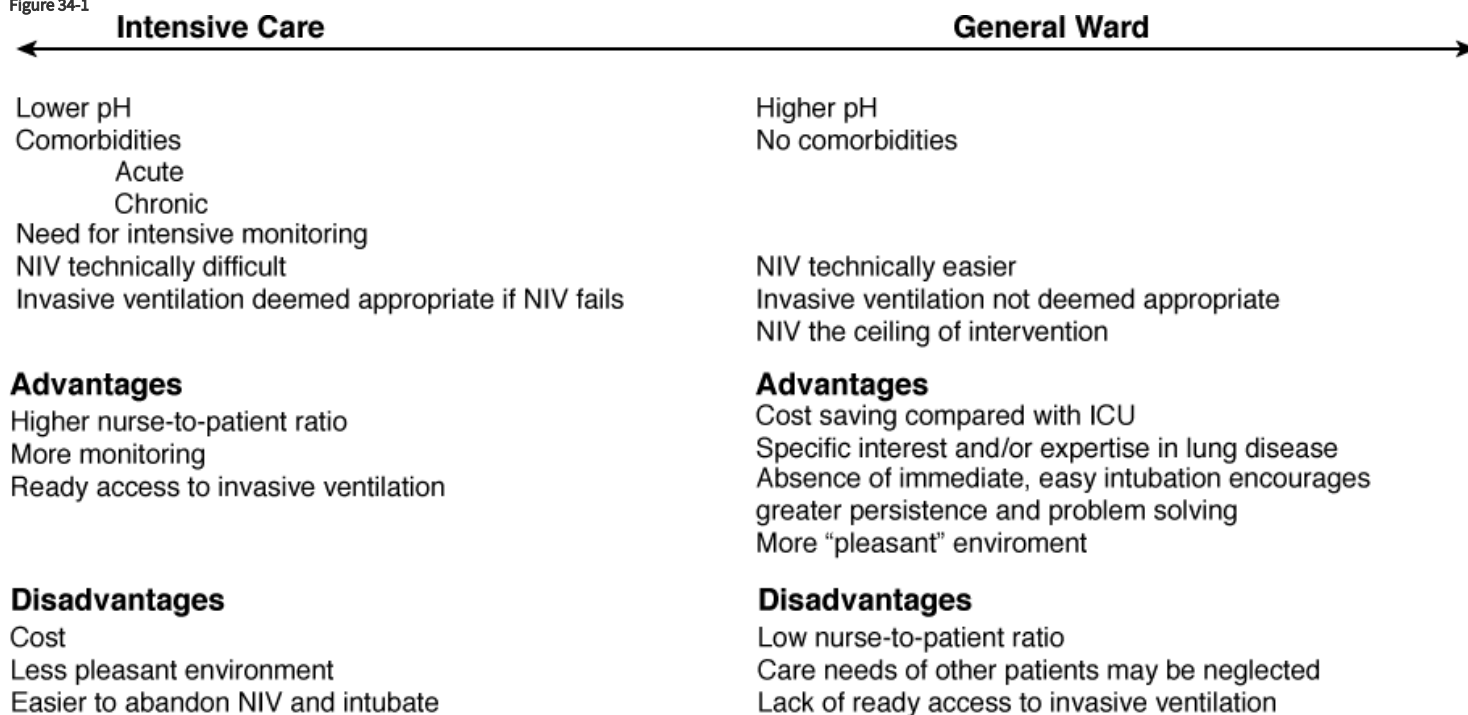
Where Should Noninvasive Ventilation Be Performed?

Acute Noninvasive Ventilation: ICU or General Ward?

There have been no direct comparisons of outcome with NIV delivered in the ICU, in intermediate units, and in a general ward. It should be appreciated that while there is some overlap, the skills needed for NIV are different from those required for invasive ventilation. Familiarity with and confidence in NIV by all members of the multidisciplinary team is the most important factor. Nurses, physiotherapists, or respiratory therapists may be the primary caregiver; this will depend on local availability, enthusiasm, and expertise. The outcome from NIV is likely to be better on a general ward where the staff has a lot of experience of NIV than in an ICU with high nurse-to-patient, therapist-to-patient, and doctor-to-patient ratios, and a high level of monitoring, but little experience of NIV. The less intensive atmosphere of a noninvasive unit may not be as distressing for patients and their relatives. NIV may be quite time-consuming in the early stages, and patients may benefit from extra attention, more likely in an ICU compared with less-well-staffed areas. Staffing is usually less at night, but a study in an ICU revealed no difference between patients failing NIV during the day and the night⁴⁹; because of the greater number of patients under the care of an individual nurse, this may not be true on a general ward and should be evaluated further.

Assuming that the skills to deliver NIV are equal in the various possible locations, there are a number of other factors to be considered. These include whether or not intubation is considered appropriate should NIV fail, the presence of other system failure, comorbidity, severity of the respiratory failure, and likelihood of success with NIV ([Fig. 34-1](#)). Patients who cannot sustain ventilation for more than a few minutes when acutely unwell require continuous observation. This level of support is more likely to be available in the ICU than in other ward environments.

Figure 34-1



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
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The spectrum of provision for an acute NIV service.

Although there are no published data, anecdotally there may be a tendency to abandon NIV more readily in an ICU because intubation is easily available, and in some ways it is easier for staff to manage a paralyzed, sedated patient than one who is struggling with NIV. When intubation is not immediately an option, there is a need to keep going a little longer, and a number of patients who at first sight appear to be failing can be managed successfully with persistence. It certainly has been the experience of the author that problems have been solved between the time that the ICU staff has been contacted and its arrival in the ward to intubate and/or transfer the patient (10 to 15 minutes).

Another important difference between the ICU and the general ward is the complexity of monitoring and the types of ventilators available. Monitoring serves two roles: (a) safety and (b) optimization of ventilator settings. ICU ventilators differ from the portable devices designed primarily for home use but frequently used in general wards. The principal limitation to the use of home ventilators during acute respiratory failure is the lack of direct online monitoring of pressure, volume, and flow provided by these devices. The evaluation of patient-ventilator asynchrony is easier with visualization of flow and pressure waveforms.⁵⁰ This may be important, particularly during the initiation of ventilation, when it is important to assess patient-ventilator interaction, respiratory mechanics, and the expired tidal volume.⁵¹ Further work is needed to establish which variables should be monitored to optimize NIV. It should be appreciated that high-technology monitoring is never a substitute for good clinical observation.⁵² For safety, it is recommended that all patients receiving NIV for acute ventilatory failure should have continuous monitoring of oxygen saturation (SO_2) by pulse oximetry, regular assessments of arterial blood-gas tensions (because there is no accurate and reliable noninvasive measure of P_{CO_2} or, more importantly, of pH), and respiratory rate.^{53,54} The SO_2 should be maintained at around 88% to 92%^{55,56} to avoid the twin dangers of dangerous hypoxia and the risk of worsening hypercapnia secondary to altering the dead-space-to-tidal-volume ratio.⁵⁷ There is no reason why this level of monitoring cannot be provided in a general ward (Table 34-2).

Table 34-2: Monitoring during Noninvasive Ventilation

Essential

- Regular clinical observation
- Continuous pulse oximetry
- Arterial blood gases after 1 to 4 hours of NIV and after 1 hour of any change in ventilator settings or fractional inspired oxygen concentration (FI_{O_2})
- Respiratory rate—continuous or intermittent

Desirable

- Electrocardiogram
- More detailed physiologic information such as leak, expired tidal volume, and measure of ventilator-patient asynchrony

If NIV is only to be provided in the ICU, the number of patients needing ICU care will increase, and this may not be necessary or appropriate. The study of Plant et al²² showed that NIV is an option outside the ICU, but the outcome for patients with a pH of less than 7.30 was not as good as that seen for comparable patients in the studies performed in a higher-dependency setting. Also, for reasons of training, throughput, quality of service, and skill retention, NIV is best performed in a single location.²⁴ An intermediate unit with ready access to an ICU is probably the best compromise.⁵⁸ A study³ of 756 consecutive patients admitted to twenty-six respiratory intermediate care units in Italy showed a better outcome than that expected on the basis of Acute Physiology and Chronic Health Evaluation (APACHE) II scores. The predicted inpatient mortality was 22.1%, whereas the actual mortality was 16%. All but forty-eight patients had chronic respiratory disease, mainly COPD ($n = 451$).

Patients with acute CPE should usually receive ventilator support where they are, because by the time arrangements have been made and transfer effected, most patients will have either improved or deteriorated to the point at which intubation is needed. Sufficient patients will attend the emergency room with CPE or develop it in the coronary care unit to make staff training in NIV in these areas worthwhile and to ensure enough throughput to maintain skills. When patients develop CPE outside these areas (e.g., in a surgical ward), it is likely to be sufficiently infrequent that it is not worth training the staff, or if trained, staff will not have had sufficient

opportunity to use and develop their skills. This is a situation in which the peripatetic NIV team or critical care outreach service may have a role. There may also be a role for CPAP delivered by helmet as it is easier to train staff in its use; it has been used successfully in the prehospital setting.⁵⁹ It is important that personnel are able to recognize and treat arrhythmias and myocardial ischemia.⁶⁰

Elective Noninvasive Ventilation: General Ward or Chronic Care Facility?

The onset of established chronic ventilatory failure is usually insidious. Patients at risk are best seen and regularly reviewed in specialist centers so that the onset of significant nocturnal hypoventilation and the development of diurnal ventilatory failure can be anticipated. As a result, NIV usually should be instituted before the patient becomes critically ill. It is advisable to acclimatize patients to NIV at an early stage once the development of significant ventilatory failure becomes likely. In one study, 90% of patients without daytime hypercapnia but with a rise in transcutaneous CO₂ during sleep required NIV within 1 year.⁶¹ Decompensation may occur with an intercurrent event, most commonly respiratory tract infection. NIV is much easier in the acute situation if the patient has experienced it previously when reasonably well. In such patients, if they are clinically stable, there is no need for assisted ventilation to “work” immediately; indeed, it does not matter if the patient is hardly able to use the ventilator at all initially.

There is little reason to admit these patients to the ICU, and it is questionable whether, with appropriate teams in place, the patient even needs admission to hospital. Similar outcomes were seen in an RCT of inpatient versus outpatient initiation of home ventilation in patients with neuromuscular disease and chest wall deformity. The mean (SD) inpatient stay was 3.8 (1.0) days, and the outpatient attendance sessions 1.2 (0.4). Health care professional contact time, including telephone calls, was: inpatient 177⁹⁹ minutes versus outpatient 188⁶⁰ minutes (p = NS). Two-month ventilator compliance was: inpatient 4.32 hours per night⁷ versus outpatient 3.92⁸ (p = NS) hours per night.⁶²

Some patients, particularly those with severe neuromuscular disease, have complex nursing needs, and their home environment may be better adapted to their needs than a hospital ward. It is more pleasant for relatives and caregivers to stay in their own home. Particular issues for that patient concerning positioning of equipment, ensuring adequate access to electrical power, and so on can be addressed.⁶³

If the patient is to be admitted to hospital, the choice may be between a chronic care facility and a general ward. As for acute NIV, staff expertise is the most important factor determining the best location (Table 34-3). Staffing and expertise being equal, advantages of the general ward include access to the ICU if things go wrong, and more ready access to other specialist teams because some of these patients have other complex needs, which the need to start NIV brings into focus. These advantages, however, are generalizations; depending on the nature of the chronic care facility, it may be better suited than a general ward for providing the care and support for the other needs of patients. Regardless of location, adequate control of nocturnal hypoventilation needs to be confirmed. For some patients, overnight oximetry may suffice; for others, particularly those receiving supplemental oxygen, monitoring of P_{CO2} is necessary. Patients also will need intermittent arterial blood-gas analyses. Increasingly, it is possible to interrogate, including remotely, the home NIV ventilator, which can give important insights into why ventilation is poorly tolerated or suboptimal? Many days' data are recorded, important in a technique with which problems may be intermittent and subject to night-to-night variation. More detailed respiratory variable monitoring, including chest wall and abdominal motion, may provide important insights,⁶⁴ but is not of itself a reason for admission to hospital.

Table 34-3: Elective Ventilation for Chronic Ventilatory Failure

There is likely to be great local variation. Factors that determine the best location

- Enthusiastic and trained staff
- Possibility of colocating with acute NIV unit
- Access to expertise in the management of nonrespiratory aspects of care
- Diagnostics, e.g., sleep laboratory
- Access to ICU

The Advantage of the General Ward

An acute and a chronic NIV service depends critically on local factors, particularly the skill levels of doctors, nurses, and therapists. The major advantage of the general ward is that it sits in the middle of the spectrum of locations for NIV provision and is likely to treat the greatest number of patients. Use of skills is a key factor in developing and retaining them. The skills learned looking after patients needing NIV acutely are equally relevant for patients being started electively on NIV. Familiarity with the nonrespiratory needs of patients with complex neuromuscular or musculoskeletal disorders, learned when patients are admitted electively to start NIV, are transferable to the care of such patients needing NIV acutely or for weaning. Continuity of care is also important. Some patients start home ventilation after an acute event, and the option of dealing with both aspects in the same place and with the same care team probably is advantageous to the patient and caregivers. If NIV is the ceiling of treatment, admission to an ICU is not necessary; if NIV is failing, the patient may be allowed to die in a less high-tech environment than that afforded by most ICUs.

There is an emerging role for NIV in the management of patients with end-stage COPD. It is well recognized that these patients receive suboptimal palliative care toward the end of life⁶⁵; in part, this relates to the great difficulty in recognizing when the end is near. The boundary between life-sustaining therapy and palliation can move on a daily basis during an exacerbation as a patient's condition improves and then deteriorates again.⁶⁶ One of the biggest challenges facing doctors now and in the future is knowing when "enough is enough."⁶⁷ NIV provides a useful alternative to life-sustaining therapy, with limited palliative care, that is, invasive ventilation, or to treatment that may not be sufficient to sustain life, but does allow effective palliative care, that is, medical therapy alone. With NIV, patients retain a real say in their care, and, because assisted ventilation is not all or nothing, it is possible to move relatively easily between life sustaining and palliative care as the patient's situation changes.

Selection of Patients for Noninvasive Ventilation in a General Ward

Acute Respiratory Failure

Chronic Obstructive Pulmonary Disease

The pH at the time NIV is initiated is the best single predictor of severity and the likelihood of success with the noninvasive approach.⁶⁸ Moreover, changes in pH and respiratory rate are easily measurable and useful in predicting the likelihood of a successful outcome from NIV.^{9,17,69–73} Arterial blood gases should be checked at baseline and after 1 to 4 hours. Data from the largest study⁷⁴ showed that hydrogen ion concentration at enrollment (odds ratio 1.22 per nmol/L) and PaCO_2 (odds ratio 1.14 per kPa) were associated with treatment failure. After 4 hours of therapy, improvement in acidosis (odds ratio 0.89 per nmol/L) and/or fall in respiratory rate (odds ratio 0.92 per breath per minute) were associated with success. If at least one of these two variables was improving, successful NIV was likely. pH, therefore, is useful in determining, first, who should receive NIV, second, in what location, and, finally, when the patient can move to a more or less intensive location. Generally speaking, the lower the pH, the greater is the risk to the patient of needing invasive ventilation if NIV is not offered or, if it is attempted, of failure. The more acidotic the patient, the greater is the need for that patient to be managed in an ICU because the risk of failure of NIV and the potential need for endotracheal intubation is higher.⁷⁵

The same criteria can be extended to other patients with acute-on-chronic hypercapnic respiratory failure, for instance those with obesity. The pH criterion is less important as some of these patients will require chronic ventilator support, with either CPAP or NIV, if they already have symptoms of excessive daytime sleepiness or peripheral edema before admission. The patient should be reassessed post discharge because a significant proportion will be able to change their mode of ventilation or even need no ongoing ventilator support.²⁹ NIV certainly should not be delayed until acidosis occurs in patients with neuromuscular disease; by the time these patients become hypercapnic, respiratory reserve is very reduced and they are at high risk. NIV should be considered even if the patient is normocapnic if he or she is also tachypneic; these patients will tire out and the PaCO₂ will start to rise. NIV will prevent this and also offload the respiratory muscles providing relief from dyspnea. Although it is less clear cut, the same principles apply to patients with chest wall deformity; chronic hypercapnia is an indication to start domiciliary NIV,⁷⁶ and there is therefore no reason not to start NIV if the patient is admitted to hospital with respiratory disease, even if not accompanied by acidosis. These patients do not require admission to an ICU.

Hypoxemic Respiratory Failure

These patients are best managed in an ICU because the risks of failure are higher and because the major problem is inadequate oxygenation. Patients are more likely to need prompt invasive ventilation if they are deteriorating or have other organ failure; moreover, ventilators usually used on general wards are those designed primarily for home use, and a high FiO₂ cannot be delivered. One further consideration was highlighted by Delclaux et al⁷⁷ in a study on the use of noninvasive CPAP in patients with hypoxemic respiratory failure; there was a trend toward more cardiorespiratory arrests in the CPAP group. The increase was attributed to the improvement in oxygenation and other physiologic parameters while the patients were using CPAP, which led to a false sense of security; when patients take the mask off, even for a short period, SO₂ may fall rapidly, putting them at high risk. Any patient who desaturates within seconds of removing a mask should be monitored very carefully, usually in an ICU, and probably this should be considered an indication for intubation.

Implications for Staffing and Training

Table 34-4 lists key training requirements.

Table 34-4: Key Training Requirements

Understanding the rationale for assisted ventilation
Mask and headgear selection and fitting
Ventilator circuit assembly
Theory of operation and adjusting ventilation to achieve desired outcome
Principles and practice of humidification
Inhaled therapy for the patient receiving NIV
Cleaning and general maintenance
Understanding how to monitor progress
Ethical issues relevant to the care of patients with incurable disease
Problem solving—the ability to recognize serious situations and act accordingly

NIV has been reported to be a time-consuming procedure.⁷⁸ As with any new technique, there is a learning curve, and the same authors subsequently published more encouraging results.⁷⁹ A number of ICU studies have shown that a significant amount of time is required to establish the patient on NIV, but this drops off substantially in subsequent days.^{10,80,81} It is possible, therefore, that NIV may have a much greater impact on nursing workload outside the ICU, where nurses have responsibility for a larger number of patients. In the study of Plant et al,²² NIV resulted in a modest increase in nursing workload, assessed using an end-of-bed log, in the first 8 hours of the admission, equivalent to 26 minutes, but no difference was identified thereafter. No data exist, however, on the effect NIV on the care of

other patients on the ward, nor whether outcome would have been better had nurses spent more time with patients receiving NIV. Most of the centers that participated in the study had little or no previous experience of NIV and therefore required training in mask fitting and application of NIV. Formal training in the first 3 months of opening a ward by a research doctor and nurse was 7.6 hours (SD 3.6). Thereafter, each center received 0.9 hour per month (SD 0.82) to maintain skills. It should be appreciated that there was no need to make subtle adjustments to ventilator settings, which all was done according to protocol. Much more training would be needed if sophisticated ventilators are used. This underlines the fact, however, that NIV, in whatever location, is not just a question of purchasing the necessary equipment but also of staff training.

Although considerable input is likely when a unit commences to provide an NIV service, thereafter, as long as a critical mass remain, new staff will gain the necessary skills from their colleagues. Given that NIV in the more severely ill patient may require as much input as an invasively ventilated patient,⁸¹ there usually should be one nurse responsible for no more than three or four patients, although this will depend on the other care needs of the patients. In the less severely affected patient, NIV can be successful with a lower level of staffing.²²

Economic Considerations

Although the findings are not consistent, some of the larger studies show that NIV can shorten length of an ICU and/or hospital stay compared, for example, with medical therapy or invasive ventilation.^{9,10,12,13,48,82} In no study has NIV been shown to lengthen hospital stay. Although not the primary aim of the studies, the finding of reduced length of stay creates or saves resources and thereby indicates a cost benefit from NIV.

In a North American cost-effectiveness analysis⁸³ in which NIV was delivered in the ICU, the authors concluded that NIV was more effective than standard treatment in reducing hospital mortality and also less expensive, with a cost saving of about \$2500 per patient admission. Intensive care is also expensive care. Intermediate units provide an alternative to the classic ICU at reduced cost.⁵⁸ The daily costs of a ventilated patient may be reduced by two-thirds when NIV is performed in a specialized respiratory unit rather than in an ICU.⁸⁴ These costs can be reduced still further when NIV is performed on a general ward,⁸⁵ although effectiveness may not be as good as in higher-dependency settings. Carlucci et al⁸⁶ showed that over time practice changes. They found that after a few years, patients with more severe acidosis were ventilated successfully with NIV; more patients received NIV on general wards, with significant cost savings, compared with when they first started providing an acute NIV service.

The Future and Important Unknowns

As the population ages, the pressure on ICU beds will increase and alternatives to intensive care will need to be developed further, even in health care systems that currently enjoy high levels of ICU provision. It is likely that, as has been seen in COPD, the management of more patients with NIV will take place outside of the ICU. The challenge will be to ensure that standards are maintained and that the real-world experience in the United Kingdom is avoided.²⁵ Technology will play a part, but staff training will remain key. The effect of provision of NIV upon the care of other patients on the general ward is an important consideration.

Conclusion

Staff training and experience are more important than location. Adequate numbers of staff, skilled in NIV, must be available throughout 24 hours. Because of the demands of looking after acutely ill patients, and to aid training and skill retention, acute NIV usually is best carried out in one single-sex location with one nurse responsible for three to four patients. Basic monitoring, at least pulse oximetry and facilities for arterial blood-gas analysis, should be available. Because the skills, both for NIV and for the other care needs of the patient population likely to need NIV, are transferable, there are significant advantages to locating both the acute and elective NIV service in the same place, but there must be ready access to invasive ventilation. The best location for both an acute and chronic NIV service will vary from institution to institution, and local expertise, enthusiasm, and hospital geography will be the major determinants of where the service should be located and how it is delivered.

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